

FONTYS UNIVERSITY OF APPLIED SCIENCES

Building a Trusted Data Space

From FIWARE Validation to Eclipse Dataspace Components
By Team Pink Panther

Phase 2A Complete | Phase 2B Initiated | March 2026

Our Journey

Three phases,
one clear mission:
build a trusted,
sovereign
data space.

01 Foundation

FIWARE DSC deployed on K3s. Components validated, infrastructure proven.

02 Validation

Use cases UC-F1 to UC-F4 confirmed working. Blockers identified for E2E.

03 Evolution

Transition to EDC for true data sovereignty, contract negotiation & Gaia-X alignment.

What We Built & Verified

SUCCESS	SUCCESS	SUCCESS	SUCCESS	PARTIAL
UC-F1	UC-F2	UC-F3	UC-F4	UC-F5
Data Broker	Identity (OID4VC)	Policy Engine	Marketplace	E2E Handshake
Scorpio NGSI-LD	Keycloak VC Issuance	ODRL → OPA/Rego	TM Forum Product Catalog	APISIX Gateway

✓ All standalone components verified operational. UC-F5 blocked by missing Verifier route mappings (/services/data-service/token) & JWS-signed VP requirement — infrastructure remediated, wallet (e.g. Walt.id) needed.

Why isn't FIWARE enough?

FIWARE Limitations

- ✗ Centralized broker model — single point of control
- ✗ No DSP-native contract negotiation — uses TM Forum APIs instead
- ✗ E2E flow blocked by SIOPv2/OIDC4VP Verifier config & JWS signing
- ✗ Aligned with DSBA/Smart City standards, not IDSA DSP

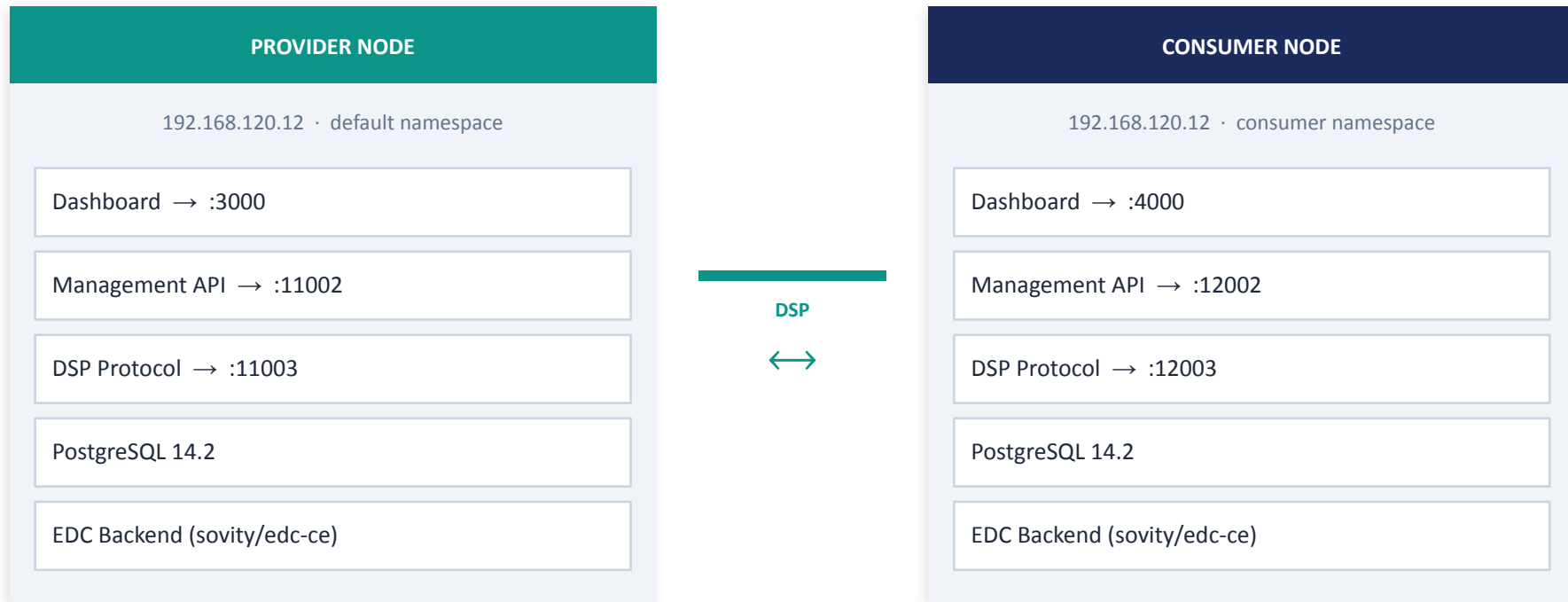
EDC Enables

- ✓ Peer-to-peer decentralized data exchange
- ✓ Automated contract negotiation (DSP protocol)
- ✓ Usage policies: time, purpose, role constraints
- ✓ Native Gaia-X & IDSA standard compliance

FIWARE vs Eclipse Dataspace Components

Feature	FIWARE (DSC)	EDC (Eclipse)	Why It Matters
Primary Goal	Real-time context mgmt	Sovereign data exchange	EDC built for cross-org trust
Architecture	Centralized broker	Decentralized P2P	No single point of failure
Policy Engine	ODRL → Rego → OPA	Full contract negotiation	Automated legal agreements
Credential Model	SIOPv2 / OIDC4VP	DCP (Decentralized Claims Protocol)	Seamless identity verification
Standards	FIWARE ecosystem	Gaia-X + IDSA global	European interoperability
Maturity for EU DS	Production (Smart City); complex to deploy	Production-ready	Scale beyond the lab

Sovity EDC Community Edition — Live on Netlab



✓ Both nodes running — data transfer verified

DATA SOVEREIGNTY

The paradigm shift
that changes
everything.

EDC transforms data
sharing from a technical
act into a contractual,
audit-ready transaction.

For Data Providers

Strict usage auditing and 'right to be forgotten' enforcement. Sensitive data can be shared safely — the policy engine prevents leakage.

For Data Consumers

Clear licensing terms and verified data streams. No legal ambiguity, no integration friction — just trusted, contract-bound data.

For the Project

Alignment with Gaia-X and IDSA standards ensures future-proofing and eligibility for large-scale EU digital initiatives.

The EDC Contract Negotiation Flow



FIWARE Context

In FIWARE, steps 1–3 worked perfectly. Steps 4–5 were blocked by missing Verifier route mappings and the requirement for a JWS-signed Verifiable Presentation from a hardware/software wallet. EDC automates this entire flow natively via the Dataspace Protocol (DSP).

Verified Assets for Phase 2B Integration

Consumer Identity (DID)

did:web:fancy-marketplace.biz

Established in FIWARE phase. Use when configuring EDC Identity Hub.

Live Data Endpoint

Scorpio Broker · 10.43.251.239:8080

EnergyReport entities ready. Map to EDC Data Address for pipeline continuity.

Network Entry Point

192.168.120.15.nip.io (Fontys Netlab)

Primary routing domain. EDC internal DNS must resolve this address.

Sovity EDC (Phase 2B)

Provider :11003 · Consumer :12003

DSP endpoints active. Both dashboards accessible on Netlab VM.

NEXT STEPS

The Road Ahead

IMMEDIATE

Connect Sovity EDC Provider to FIWARE Scorpio data stream

Phase 2B

SHORT-TERM

Implement full DSP contract negotiation flow between both EDC nodes

Phase 2B

MID-TERM

Connect Sovity EDC Provider to a real data source (Scorpio / NGSI-LD)

Phase 3

STRATEGIC

Publish verified data offering to Gaia-X / IDSA compatible catalog

Phase 3

IN SUMMARY

We proved it. Now we scale it.

Phase 2A demonstrated that every critical FIWARE component is capable of fulfilling its data-sharing role. Phase 2B takes those learnings and upgrades the foundation — replacing manual credential ceremonies with automated, standards-based contract negotiation via EDC.

4 / 5

Use Cases
Verified

2

EDC Nodes
Deployed

100%

Infrastructure
Remediated